Rett syndrome and MECP2
AIIMS data: an MECP2 change in 98.6% of classic Rett girls

For sales and healthcare teams. Facts summarised in our own words from published sources (links below). Treatment decisions are always the doctor's.

Indian study: Khajuria R, Gupta N, van Roozendaal KEP, Sapra S, Ghosh M, Gulati S, Smeets EEJ, Curfs LMG, Kabra M. *Spectrum of MECP2 mutations in Indian females with Rett syndrome: a large cohort study.* J Transl Genet Genom 2020;4:91-103. All India Institute of Medical Sciences, New Delhi. Open access.

Who was tested

- 72 sporadic girls with classic Rett syndrome (revised 2010 criteria)
- Sanger sequencing of MECP2 exons 2-4, then MLPA for deletions and duplications
- Symptom onset median 16 months; diagnosis median 42 months

What was found

- **98.6%** overall: sequence variants 88.9%, large deletions by MLPA 9.7%
- 38 distinct variants, 13 new; commonest p.T158M and p.R270X
- 40% could not walk at inclusion; recurrent nonsense variants linked to scoliosis

Second Indian study

Lallar et al., Indian Pediatrics 2018;55:474-7 (SGPGI Lucknow, AIIMS): MECP2 change in **93% (13/14) of classic** and 20% (1/5) of atypical Rett; **0 of 17** girls with Rett-like features who did not meet the criteria. Two classic girls had large deletions found only by MLPA. The authors conclude that the 2010 criteria guide whom to test.

Facts and diagnosis

- **Frequency:** about 1 in 10,000 to 23,000 females; one of the most common genetic causes of developmental impairment in girls. Atypical Rett pictures also occur with CDKL5 and FOXG1 (ACMG).
- **Five necessary signs (NICHD):** development, regression, then stabilisation; loss of purposeful hand use; loss of spoken language; repetitive hand movements; abnormal gait. Slowing head growth between 3 months and 4 years should prompt the diagnosis.
- A positive MECP2 test alone does not make the diagnosis; clinical assessment does (NICHD).
- **Treatment:** trofinetide received FDA accelerated approval on 10 March 2023 for Rett syndrome in patients aged 2 years and older (USA).
- Prenatal testing in a later pregnancy: disease only; fetal sex never disclosed (PCPNDT).

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Rett syndrome: how to approach doctors

Topic	What to do or say
Doctors to visit	Paediatric neurologists, developmental paediatricians, child development and therapy centres, paediatric orthopaedics (scoliosis).
Which girls	Normal early months then lost skills; stopped purposeful hand use; lost words; constant hand wringing or clapping; slowing head growth.
Opening question	"Doctor, in a girl who lost hand use and speech, do you test MECP2 with MLPA?"
Facts to quote	AIIMS: 98.6% of 72 classic Rett girls had an MECP2 change; about 1 in 10 found only by MLPA. Diagnosis about 2 years after symptoms (medians 16 and 42 months).
Test to suggest	MECP2 sequencing plus deletion/duplication (MLPA). If negative or atypical: CDKL5, FOXG1, a gene panel or exome with CNV analysis. **Not on our standard menu: confirm availability, price and report time with the lab first.**
"The diagnosis is clinical"	"Yes; the report confirms it, and about 1 in 10 at AIIMS needed MLPA to find the change."
"There is no treatment"	"A first medicine (trofinetide) was approved in the USA in 2023; the report also guides family counselling."
Sample	EDTA blood; counselling and written consent first.
Never say	That the test or any medicine will cure the child. Medicine decisions are always the doctor's.

References

- Khajuria R et al. Spectrum of MECP2 mutations in Indian females with Rett syndrome: a large cohort study. J Transl Genet Genom 2020;4:91-103. oaepublish.com/articles/jtgg.2020.06
- Lallar M et al. Molecular testing of MECP2 gene in Rett syndrome phenotypes in Indian girls. Indian Pediatr 2018;55:474-7. indianpediatrics.net/june2018/474.pdf
- NICHD. How do healthcare providers diagnose Rett syndrome? nichd.nih.gov
- ACMG. Trofinetide approved for children and adults with Rett syndrome. acmg.net
- Neul JL et al. Rett syndrome: revised diagnostic criteria and nomenclature. Ann Neurol 2010.

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For general awareness of healthcare and sales teams. Not a substitute for clinical judgement; facts and guidelines may change.